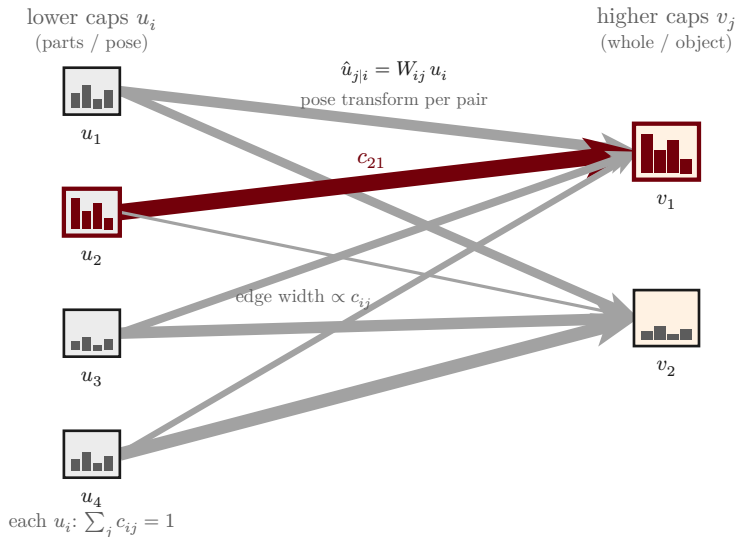


(a) routing between capsule layers



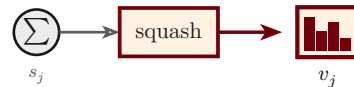
(b) forming a parent capsule v_j

vote: $\hat{u}_{j|i} = W_{ij} u_i$

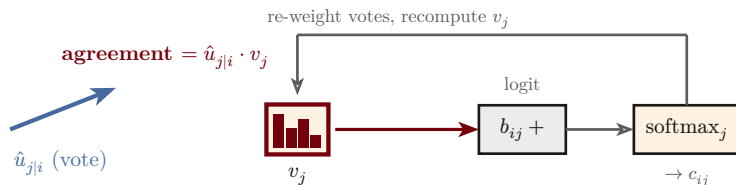
combine: $s_j = \sum_i c_{ij} \hat{u}_{j|i}$

squash (length $\rightarrow [0, 1]$, keep direction):

$$v_j = \frac{\|s_j\|^2}{1 + \|s_j\|^2} \cdot \frac{s_j}{\|s_j\|}$$



(c) routing by agreement (iterate r times)



votes that AGREE with v_j grow c_{ij} ; disagreeing votes shrink